

W01 WATER QUALITY INDICATORS | P

Intent : Verify the quality of water for human contact through easy-to-test parameters.

Summary : This WELL feature requires the provision of water that meets thresholds for turbidity and coliforms for all water likely to come in contact with building occupants and verifies performance using on-site tests.

Issue : Most cities have an extensive treatment system to produce and deliver water with safety and integrity. Two parameters—total coliforms and turbidity— are commonly used to assess the effectiveness of these systems as easy-to-test measurements for the possible presence of other, more concerning contaminants, and are therefore known as 'indicators'. Coliform bacteria are naturally present in the environment and are generally considered harmless. However, some coliforms are associated with fecal contamination and may cause disease if ingested.¹ Turbidity is a measure of water cloudiness, which per-se does not constitute a health concern, however, it does relate to the availability of food and shelter for microbes, the presence of particulate contaminants and issues with the water treatment process,² along with posing an aesthetic concern. High turbidity water also can reduce the efficacy of water treatment technologies.¹

Solutions : Water filtration can reduce turbidity and, depending on the type of device, may also trap bacteria and other contaminants. If the water has low turbidity, disinfection at the point of use with ultraviolet (UV) light may be effective at killing coliforms and pathogenic microbes.²

PART 1 VERIFY WATER QUALITY INDICATORS

For All Spaces:

Water delivered to the project and intended for human contact (e.g., drinking, cooking and dishwashing, handwashing, showering or bathing) meets the following thresholds:

- a. Turbidity is less than or equal to 1.0 NTU, FTU or FNU (nephelometric turbidity, formazin turbidity or formazin nephelometric units, respectively).
- b. Coliforms are not detected in any 100 ml sample.

Note : Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. US Centers for Disease Control & Prevention. A Guide to Drinking Water Treatment Technologies for Household Use. [Reference](#). Published 2014. Accessed December 10, 2019.